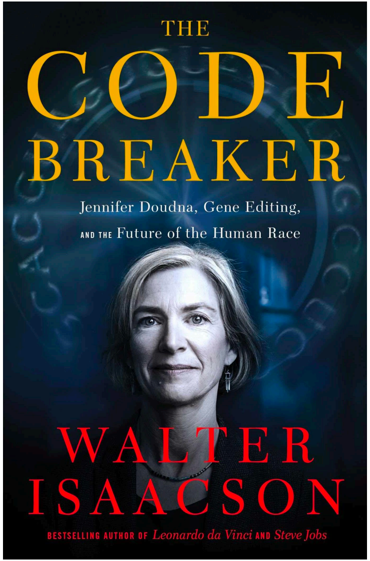
Assortment Optimization and Beyond

Jake Feldman

CRISPR Book


THE
CODE
BREAKER
Jennifer Doudna, Gene Editing,
AND THE Future of the Human Race
WALTER
ISAACSON
BESTSELLING AUTHOR OF Leonardo da Vinci AND Steve Jobs

Outline

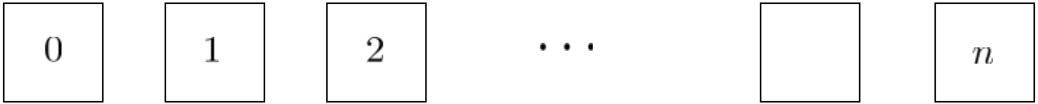
The Assortment Problem

The Deterministic LP

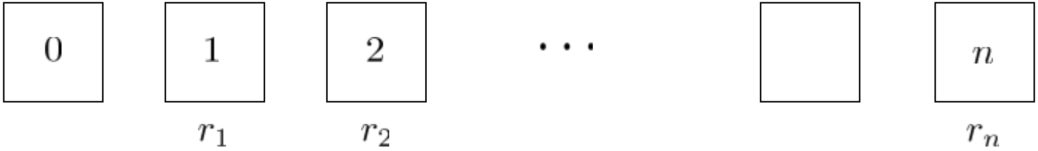
Choice-Based Online Matching

Part 1 – Assortment Optimization

The General Assortment Problem



The General Assortment Problem



The General Assortment Problem

0	1	2	$\cdots$		n
	r_1	r_2			r_n

$$\max_{S \subseteq [n]} \mathcal{R}(S)$$

$[n] = \{1, \ldots, n\}$

Expected revenue of S

The General Assortment Problem

0	1	2	$\cdots$		n
	r_1	r_2			r_n

$$\max_{S \subseteq [n]} \mathcal{R}(S) = \max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i$$

Probability i is selected under assortment S

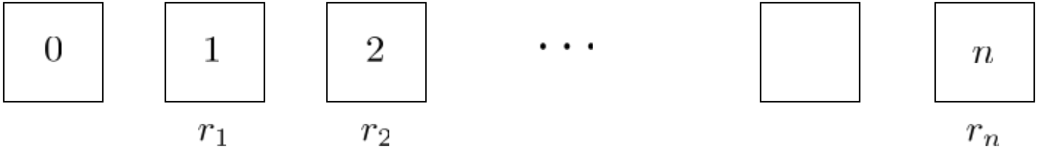
The General Assortment Problem

$$\boxed{0} \quad \underset{r_1}{\boxed{1}} \quad \underset{r_2}{\boxed{2}} \quad \cdots \quad \boxed{} \quad \underset{r_n}{\boxed{n}}$$

$$\max_{S \subseteq [n]} \mathcal{R}(S) = \max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i$$

Why is this problem “hard”?

The General Assortment Problem



$$\max_{S \subseteq [n]} \mathcal{R}(S) = \max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i$$

Independent demand model - $\pi(i, S) = \pi_i$

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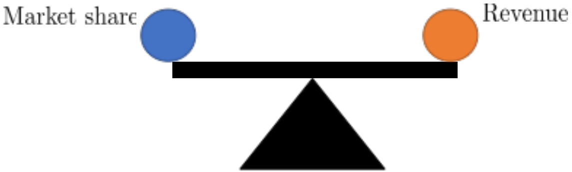

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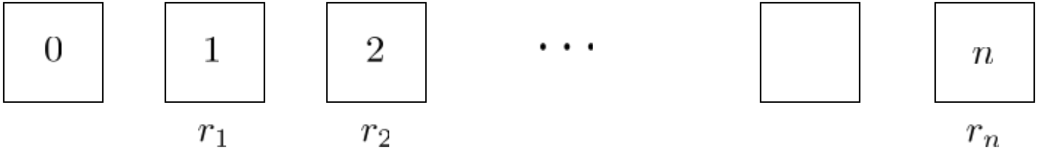

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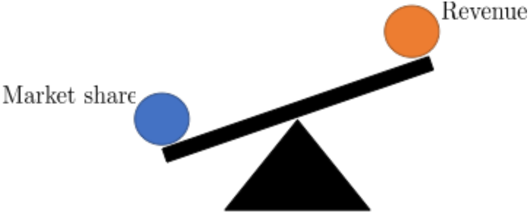

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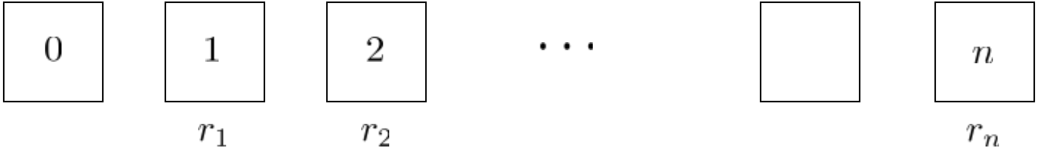

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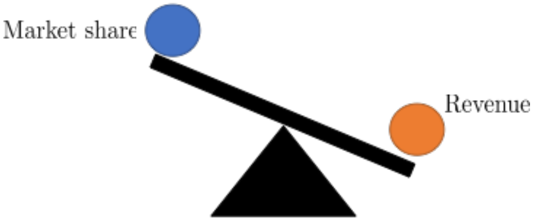

The Multinomial Logit (MNL) Model

$\boxed{0} \quad \boxed{1} \quad \boxed{2} \quad \cdots \quad \boxed{i} \quad \cdots \quad \boxed{n}$

v_i

$$\boxed{\max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i =}$$

The Multinomial Logit (MNL) Model

$$\boxed{0} \quad \boxed{1} \quad \boxed{2} \quad \cdots \quad \underset{v_i}{\boxed{i}} \quad \cdots \quad \boxed{n}$$

$$U_i = v_i + \epsilon_i$$

$$\boxed{\max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i =}$$

The Multinomial Logit (MNL) Model

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$$U_i = v_i + \epsilon_i$$

Gumbel iid component

random utility of i

deterministic component

$$\max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i =$$

The Multinomial Logit (MNL) Model

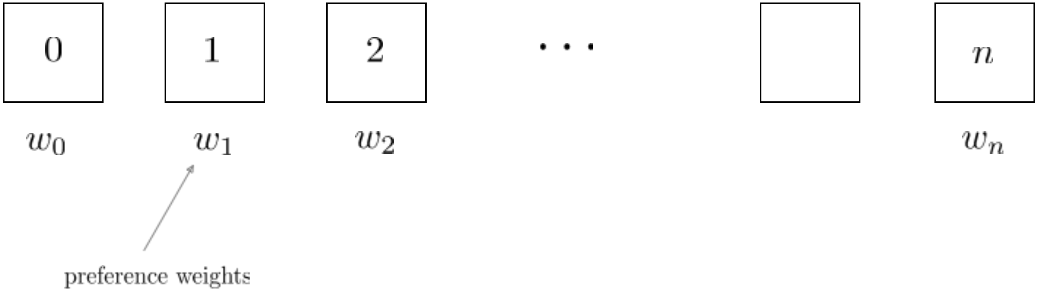
$$\boxed{0} \quad \boxed{1} \quad \boxed{2} \quad \cdots \quad \underset{v_i}{\boxed{i}} \quad \cdots \quad \boxed{n}$$

$$U_i = v_i + \epsilon_i$$

$$\pi(i, S) = \Pr\left[U_i \geq \max_{j \in S_+} U_j\right] = \frac{e^{v_i}}{1 + \sum_{j \in S} e^{v_j}} = \frac{w_i}{1 + \sum_{j \in S} w_j}$$

$$\boxed{\max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i =}$$

The Multinomial Logit (MNL) Model



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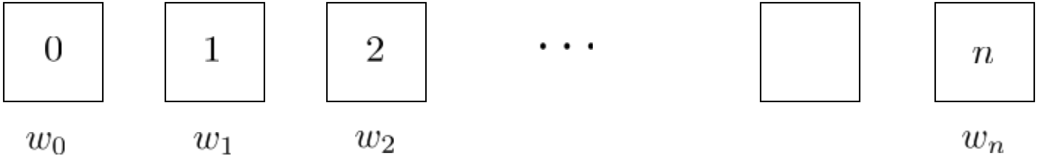

$$\pi(i, S) = \frac{w_i}{w_0 + w(S)}$$

$$w(S) = \sum_{i \in S} w_i$$

$$\max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot r_i =$$

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0	1	2	$\cdots$		n
w_0	w_1	w_2			w_n

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Kenneth E Train. *Discrete Choice Methods with Simulation*

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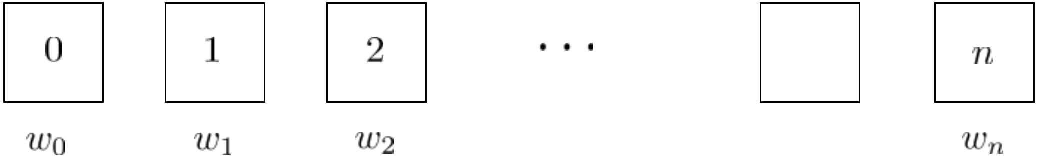

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The MNL Assortment Problem

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(S^* is the optimal assortment)

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Theorem: The optimal assortment S^* is revenue-ordered, i.e. $S^* = \{1, \ldots, k\}$ for some $k \in [n]$.

Talluri K, Van Ryzin G (2004) *Revenue management under a general discrete choice model of consumer behavior.*

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How do I find the optimal assortment?

MNL Assortment – LP Formulation

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x_i : choice probability of product i

$\hat{S} = \{i \in [n] : x_i^* > 0\}$, then $x_i^* = \pi(i, \hat{S})$

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(1) $x_0 + \sum_{i \in [n]} x_i = 1$

$\forall\, i \in [n]$

(3) $x_i \geq 0$

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Intuition for constraint (2):

For any assortment S, and product $i \in S$

$$\frac{\pi(i,S)}{\pi(0,S)} = w_i \implies \frac{x_i}{x_0} = w_i$$

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At any BFS we have:

\# tight constraints = \# decisions variables

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At any BFS we have:

tight constraints = # decisions variables

$$\left.\begin{array}{c}\text{one from (1) +}\\ n \text{ from (2) and (3)}\end{array}\right. = n + 1$$

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Plugging into constraint (1):

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Plugging into constraint (1):

$$x_0^* + x_0^* \cdot \sum_{i \in \hat{S}} w_i = 1$$

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How do we show revenue-ordered is optimal?

MNL Assortment – LP Formulation

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How do we show revenue-ordered is optimal?

Let's we knew x_0^*, then the LP becomes:

$$\max \sum_{i \in [n]} r_i x_i$$

$$(1)\ \sum_{i \in [n]} x_i = C_1$$

$$(2)\ x_i \leq C_2$$

$$(3)\ x_i \geq 0 \qquad \forall\, i \in [n]$$

MNL Assortment – LP Formulation

$$\boxed{\max_{S \subseteq [n]} \sum_{i \in S} \frac{r_i w_i}{1 + w(S)}}$$

(S^* is the optimal assortment)

Adding constraints:

Mika Sumida, Guillermo Gallego, Paat Rusmevichientong, Huseyin Topaloglu, and James Davis. *Revenue-utility tradeoff in assortment optimization under the multinomial logit model with totally unimodular constraints.* Management Science

$$\max \sum_{i \in [n]} r_i x_i$$

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Basic cardinality: $\sum_{i\in[n]} x_i \leq K$

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Basic cardinality: $\sum_{i \in [n]} x_i \leq K$

Other encodings: discrete pricing, promotion, quality consistent pricing

Beyond MNL

In what ways does the MNL model fall short of reality?

Beyond MNL

In what ways does the MNL model fall short of reality?

Other choice models

Consideration sets

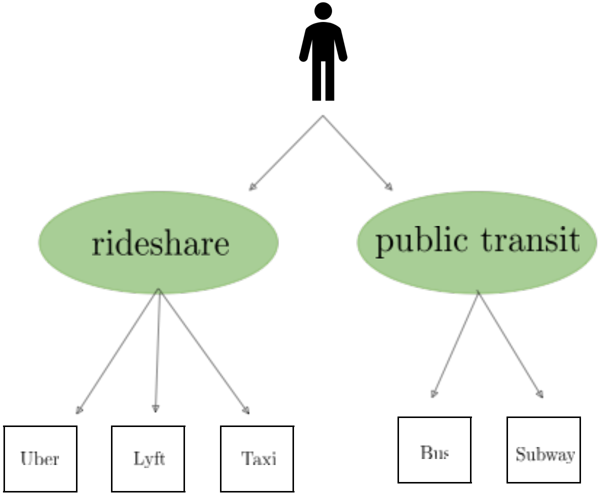
Miscellaneous

Multi-purchase

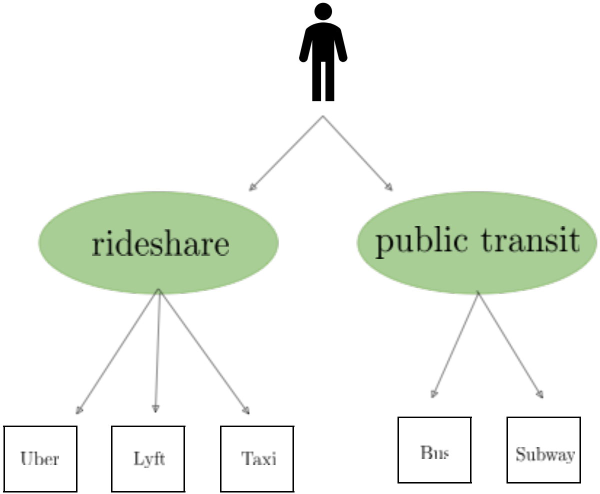
synergies

beyond revenue maximization

The Nested Logit Model



The Nested Logit Model



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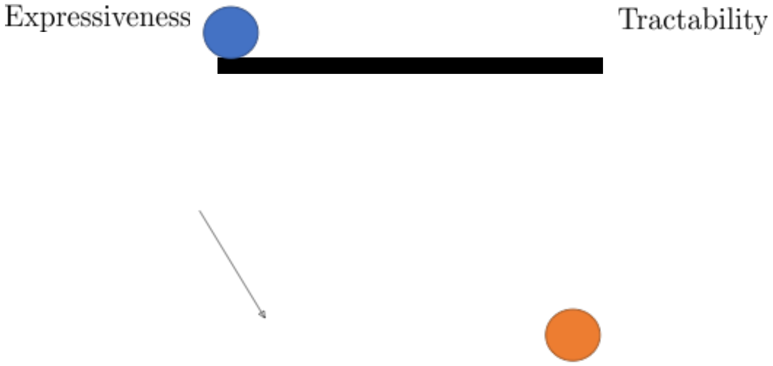
Other Choice Models



Other Choice Models

Mixed-MNL Model

Other Choice Models

Mixed-MNL Model

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Isabel Mendez-Diaz, Juan Jose Miranda-Bront, Gustavo Vulcano, and Paula Zabala. *A branch-and-cut algorithm for the latent-class logit assortment problem.* Discrete Applied Mathematics

Paat Rusmevichientong, David Shmoys, Chaoxu Tong, and Huseyin Topaloglu. *Assortment optimization under the Multinomial Logit model with random choice parameters.* Production and Operations Management

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Other Choice Models

Mixed-MNL Model

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Feldman, Aouad, Segev (2021) *The Exponomial Choice Model for Assortment Optimization: An Alternative to the MNL Model?*. Management Science

Consideration Sets

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<u>Nested Consideration Sets</u>

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Davis, James M, Huseyin Topaloglu, David P Williamson. 2015. *Assortment optimization over time*. Operations Research Letters

Gallego, Guillermo, Anran Li, Van-Anh Truong, Xinshang Wang. 2020. *Approximation algorithms for product framing and pricing*. Operations Research

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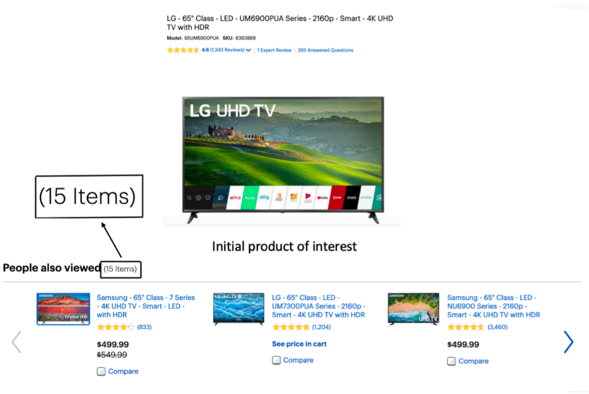
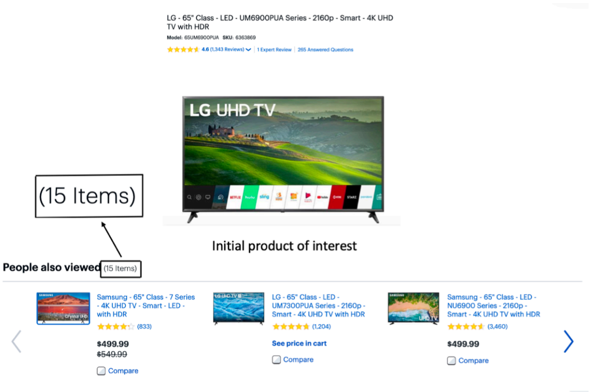
Consideration Sets

Sequential Models



5 sequential recommendation stages

Consideration Sets



Initial product of interest

5 sequential recommendation stages

Sequential Models

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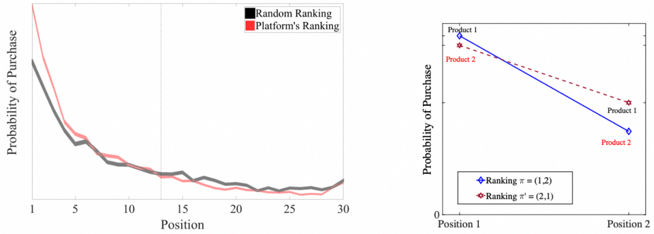
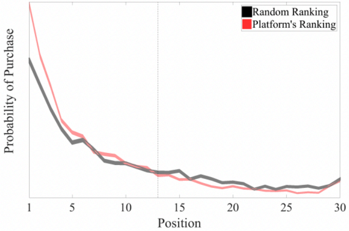
Consideration Sets

Search Cost

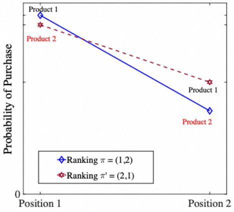


(a) (b)

Consideration Sets



(a)



(b)

Search Cost

Wang, Ruxian, Ozge Sahin. 2018. *The impact of consumer search cost on assortment planning and pricing.* Management Science

Derakhshan, Mahsa, Negin Golrezaei, Vahideh Manshadi, Vahab Mirrokni. 2022. *Product ranking on online platforms.* Management Science

Miscellaneous

Multi-purchase

Ferreira K, Goh J (2021) *Assortment rotation and the value of concealment.* Management Science

Fox E, Norman L, Semple J (2018) *Choosing an n-pack of substitutable products.* Management Science

Tulabandhula T, Sinha D, Patidar P (2020) *Multi-purchase behavior: Modeling and optimization.*

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Beyond Revenue Maximization

Mika Sumida, Guillermo Gallego, Paat Rusmevichientong, Huseyin Topaloglu, and James Davis. *Revenue-utility tradeoff in assortment optimization under the multinomial logit model with totally unimodular constraints.* Management Science

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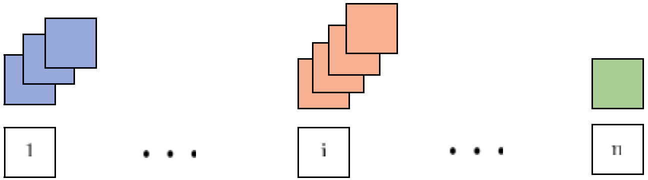
Huh, Woonghee T, Hongmin Li. 2022. *Product-line pricing with dual objective of profit and consumer surplus.* Production and Operations Management

Part 2 – Choice-Based Online Matching and the CBLP

Problem Formulation



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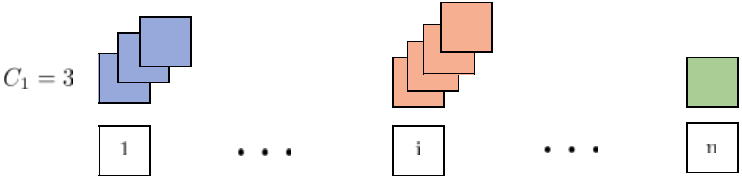

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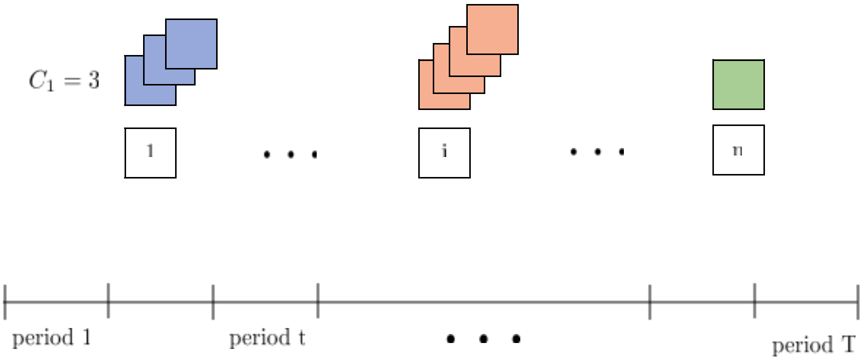

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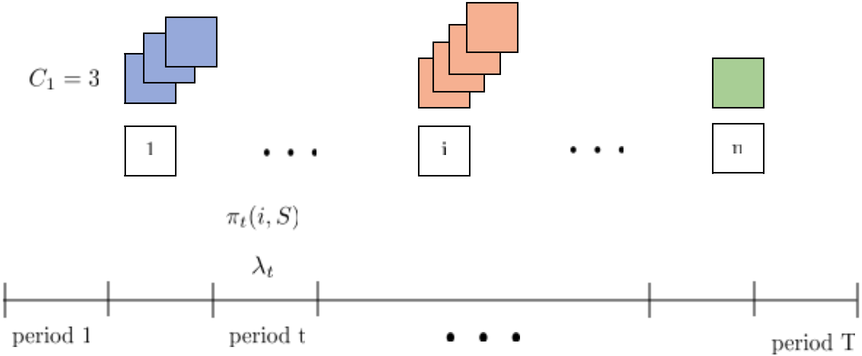
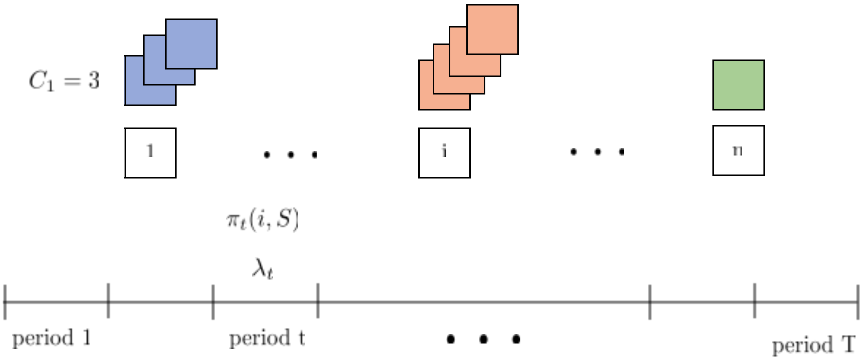


Assumption: $\pi(i, S) \geq \pi(i, Q)$ for $S \subseteq Q$

Problem Formulation

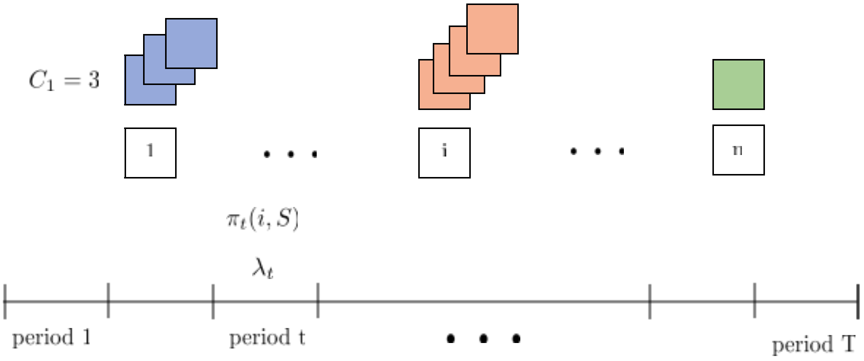


Brief interlude:

mean arrivals: $\sum_{t\in[T]} \lambda_t$

variance: $\sum_{t\in[T]} \lambda_t \cdot (1 - \lambda_t)$

Problem Formulation



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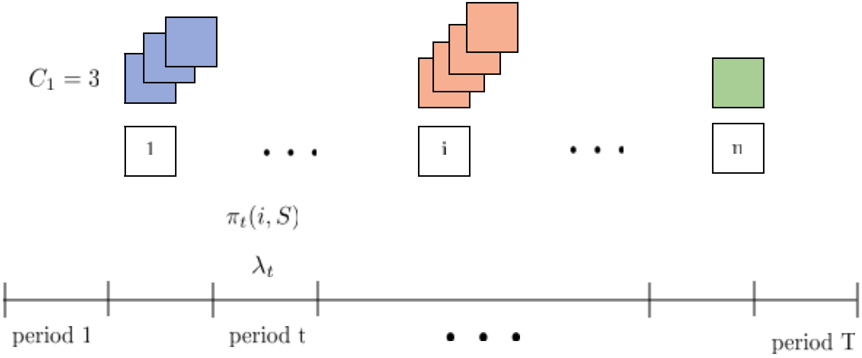
$\implies$ variance $\leq$ mean

Problem Formulation



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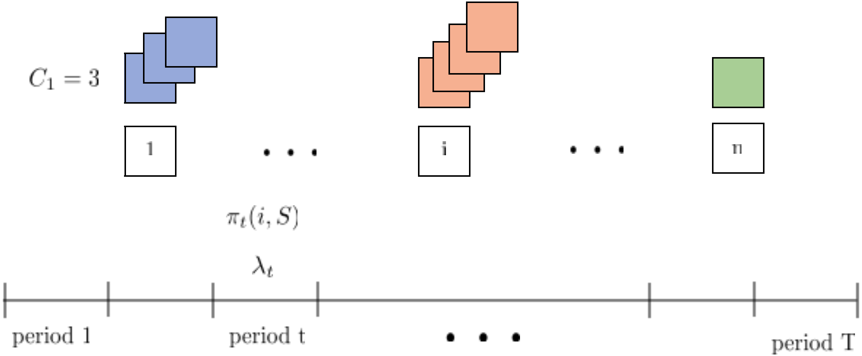
mean arrivals: $\sum_{t\in[T]} \lambda_t$

variance: $\sum_{t\in[T]} \lambda_t \cdot (1 - \lambda_t)$

$\implies$ variance $\leq$ mean

$\implies$ coef. variantion $= \frac{\sigma}{\mu} \leq \frac{1}{\sqrt{\sum_{t\in[T]} \lambda_t}}$

Problem Formulation



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Y. Bai, O. El Housni, B. Jin, P. Rusmevichientong, H. Topaloglu and D. P. Williamson, *Fluid approximations for revenue management under high-variance demand*, Management Science

Aouad, Ma, *A Nonparametric Framework for Online Stochastic Matching with Correlated Arrivals*,

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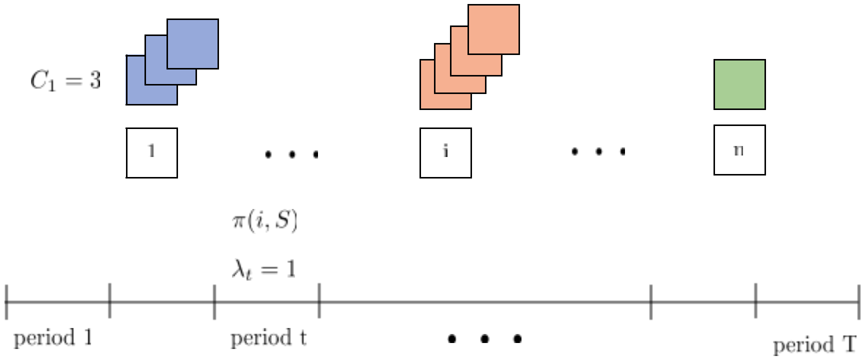

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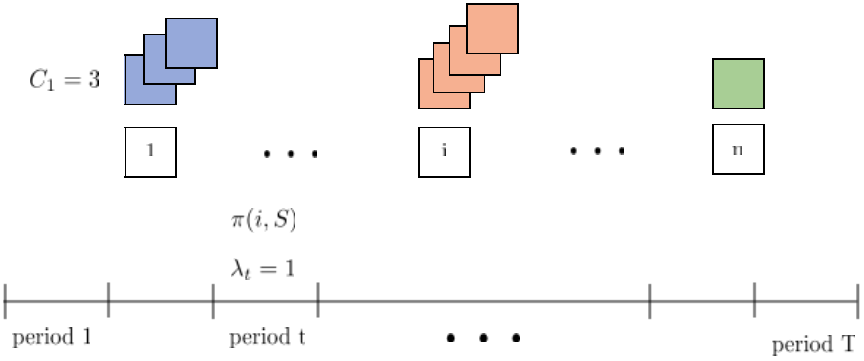
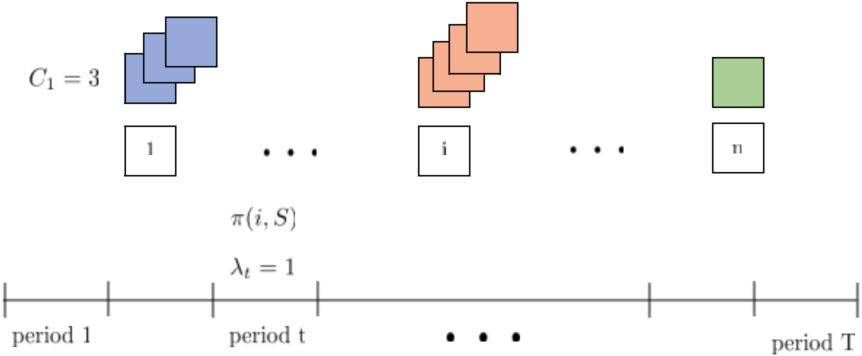


DP Formulation:

$V_t(X) =$

Problem Formulation

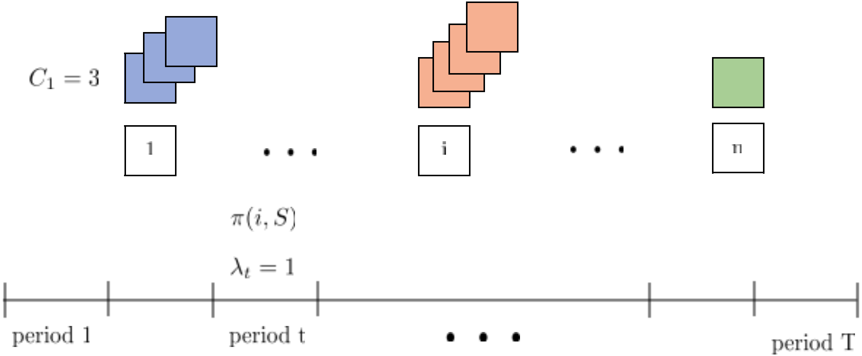


DP Formulation:

$$V_t(X) =$$

$X = (x_0, x_1, \ldots, x_n)$

Problem Formulation



DP Formulation:

$$V_t(X) = \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i + V_{t+1}(X - e_i)) + (1 - \sum_{i \in S} \pi(i,S)) \cdot V_{t+1}(X) \qquad \text{(Base cases } V_{T+1}(\cdot) = 0\text{)}$$

$$= \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i - (V_{t+1}(X) - V_{t+1}(X - e_i))) + V_{t+1}(X)$$

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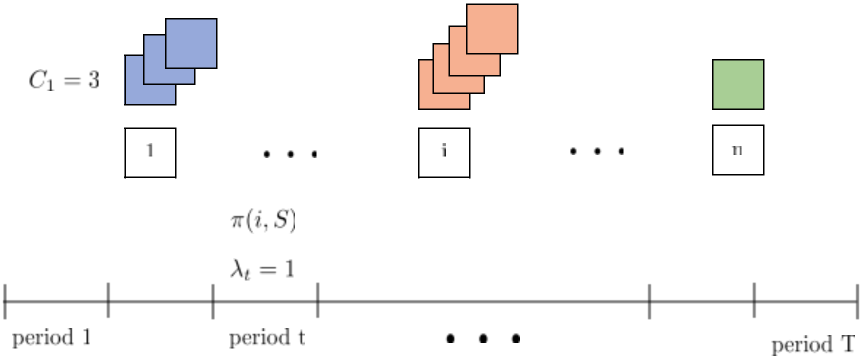

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decision variables: $h_t(S)$ - fraction of time S if offered in period t

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expected demand for product i

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Liu Q, Van Ryzin G (2008) *On the choice-based linear programming model for network revenue management.* Manufacturing and Service Operations Management

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- Can use the $h_t^*(S)$ to develop randomized policies

Theorem: A classical result is that $Z^* \geq \mathrm{OPT}$

(Later we will show that $2\mathrm{OPT} \geq Z^*$

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The Choice-Based Deterministic LP

decision variables: $h_t(S)$ - fraction of time S if offered in period t

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Why is this useful?

- Used to measure the optimality gap of heuristics
- Used compute performance guarantee of policies
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Ma W, Simchi-Levi D, Zhao J (2021) *Dynamic pricing (and assortment) under a static calendar.* Management Science

Chen, Huang, Feldman (2023) *Approximation Schemes for Dynamic Pricing with Opaque Products.*

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So, how do I solve the CBDLP??

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Dual:

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Dual:

$$\min \sum_{t\in[T]} \beta_t + \sum_{i\in[n]} C_i \alpha_i$$

$$\beta_t + \sum_{i\in S} \pi(i,S)\alpha_i \ge \sum_{i\in S} r_i \pi(i,S) \quad \forall\, t \in [T], S \subseteq [n]$$

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$$\beta_t \geq \sum_{i \in S} \pi(i, S) \cdot (r_i - \alpha_i) \quad \forall\, t \in [T], S \subseteq [n]$$

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Constraint generation:
- Start with subset of constraint, solve the LP
- Check if optimal solution violates any constraints

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Check if there exists S such: $\beta_t^* < \pi(i,S) \cdot (r_i - \alpha_i^*)$

Theorem: A classical result is that $Z^* \ge \mathrm{OPT}$

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Check: $\beta_t^* \overset{?}{\ge} \max_{S\subseteq[n]} \pi(i,S) \cdot (r_i - \alpha_i^*)$

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$$\text{Check: } \beta_t^* \overset{?}{\geq} \max_{S \subseteq [n]} \pi(i, S) \cdot (r_i - \alpha_i^*)$$

assortment problem!

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Dual:

$$\min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

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Part 3 – A Constant Factor Approximation for Choice-Based Online Matching

High-Level Idea

DP Formulation:

$$V_t(X) = \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i + V_{t+1}(X - e_i)) + (1 - \sum_{i \in S} \pi(i,S)) \cdot V_{t+1}(X) \qquad \text{(Base cases } V_{T+1}(\cdot) = 0)$$

$$= \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i - (V_{t+1}(X) - V_{t+1}(X - e_i))) + V_{t+1}(X)$$

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value of unit of product i
(at time t when inventory is X)

High-Level Idea

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value of unit of product i
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Linear value function approximation:

estimate of value of unit of product i

$$H_t(X) = \sum_{i \in [n]} \gamma_{it} x_i$$

High-Level Idea

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$$= \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i - \underbrace{(V_{t+1}(X) - V_{t+1}(X - e_i))}_{\substack{\text{value of unit of product } i \\ \text{(at time } t \text{ when inventory is } X\text{)}}}) + V_{t+1}(X)$$

Linear value function approximation:

estimate of value of unit of product i

$$H_t(X) = \sum_{i \in [n]} \gamma_{it} x_i$$

Key property: $H_t(X) - H_t(X - e_i) = \gamma_{it}$

High-Level Idea

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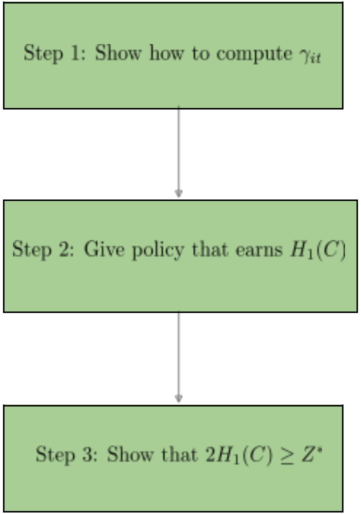
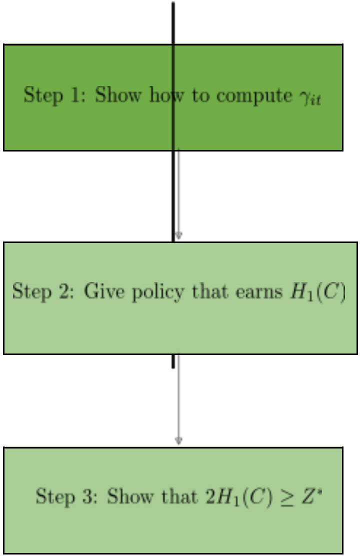
Key property: $H_t(X) - H_t(X - e_i) = \gamma_{it}$

now our estimate is inventory-independent

Outline


Step 1: Show how to compute γ_{it}
Step 2: Give policy that earns $H_1(C)$
Step 3: Show that $2H_1(C) \geq Z^*$

Outline



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Computing Tuning Parameters

γ_{it}: value of one unit of product i

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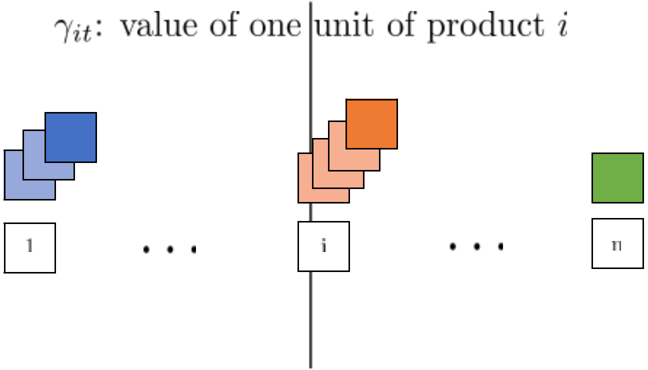

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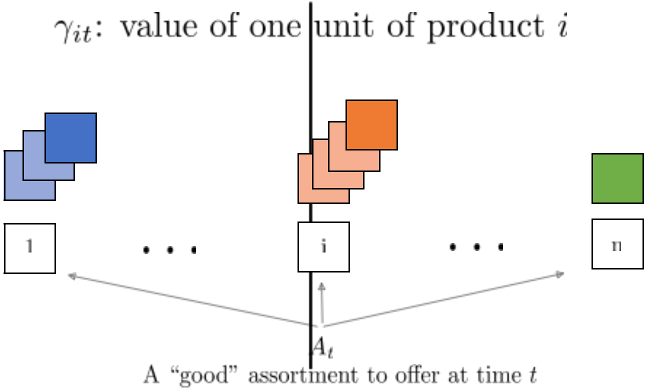

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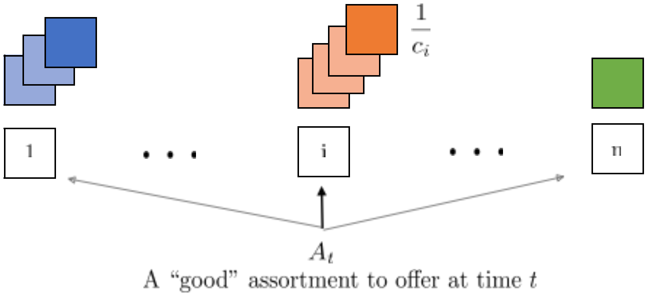

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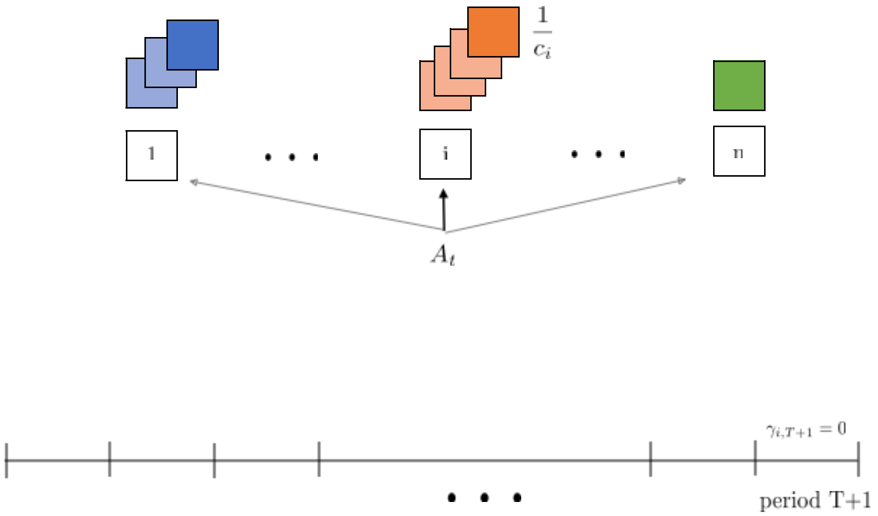

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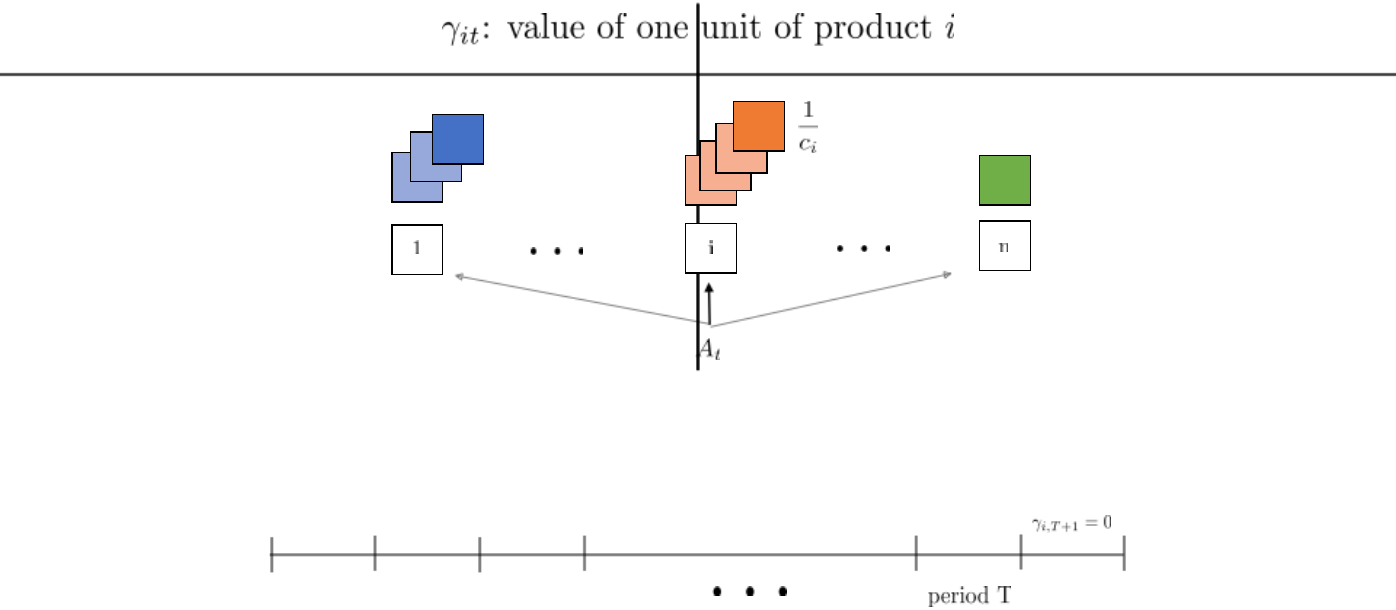

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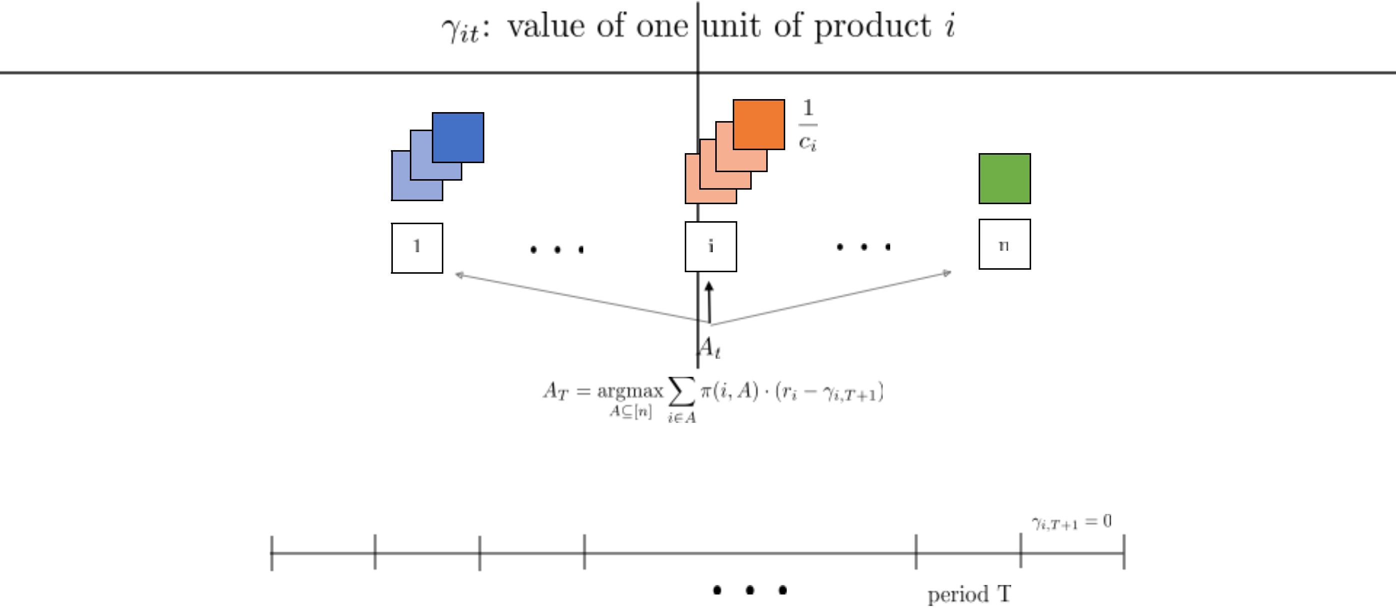

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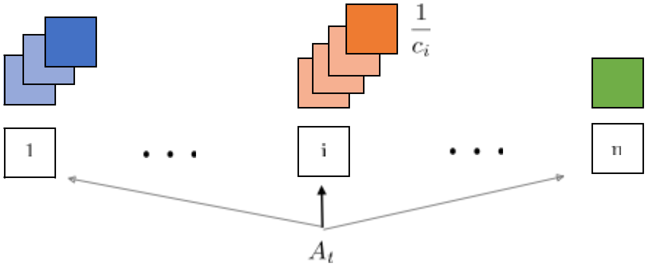
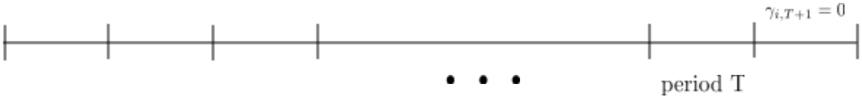


$$A_T = \operatorname*{argmax}_{A \subseteq [n]} \sum_{i \in A} \pi(i, A) \cdot (r_i - \gamma_{i,T+1})$$

$$\gamma_{iT} = \pi(i, A_T) \cdot \frac{1}{c_i} r_i + (1 - \pi(i, A_T) \cdot \frac{1}{c_i}) \cdot \gamma_{i,T+1}$$



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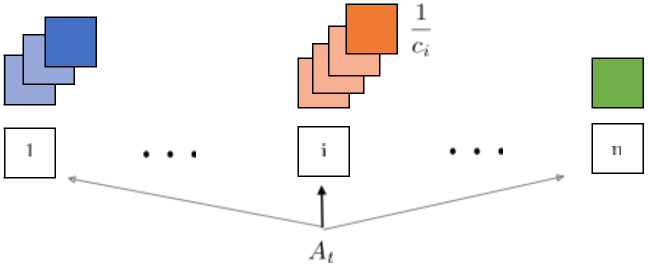

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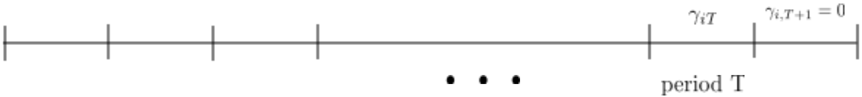

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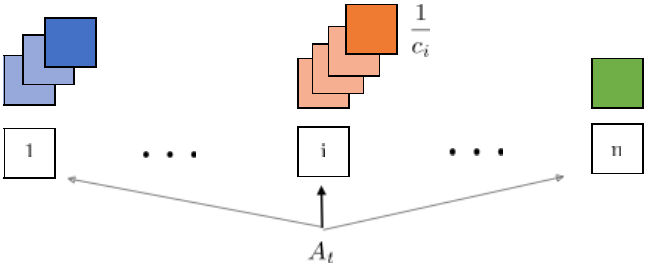
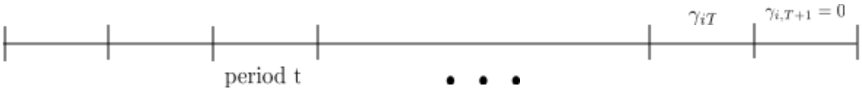


$$A_t = \operatorname*{argmax}_{A \subseteq [n]} \sum_{i \in A} \pi(i, A) \cdot (r_i - \gamma_{i,t+1})$$

$$\gamma_{it} = \pi(i, A_T) \cdot \frac{1}{c_i} \cdot (r_i - \gamma_{i,t+1}) + \gamma_{i,t+1}$$



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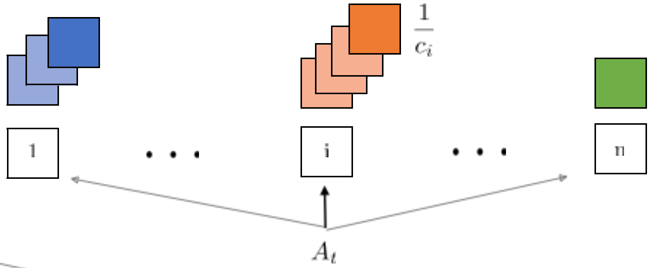

This is an assortment problem!

$$A_t = \operatorname*{argmax}_{A \subseteq [n]} \sum_{i \in A} \pi(i, A) \cdot (r_i - \gamma_{i,t+1})$$

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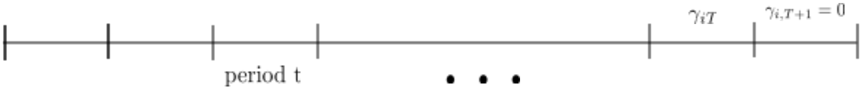

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$$\gamma_{it} = \pi(i, A_T) \cdot \frac{1}{c_i} \cdot (r_i - \gamma_{i,t+1}) + \gamma_{i,t+1}$$

Two key properties:

(i) $\gamma_{it} \geq \gamma_{i,t+1}$

(ii) $c_i \cdot (\gamma_{it} - \gamma_{i,t+1}) = \pi(i, A_t) \cdot (r_i - \gamma_{i,t+1})$

Outline

Step 1: Show how to compute γ_{it}

Step 3: Show that $2H_1(C) \geq Z^*$

The Roll-Out Policy

DP Formulation:

$$V_t(X) = \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i + V_{t+1}(X - e_i)) + (1 - \sum_{i \in S} \pi(i,S)) \cdot V_{t+1}(X) \qquad \text{(Base cases } V_{T+1}(\cdot) = 0\text{)}$$

$$= \max_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i - (V_{t+1}(X) - V_{t+1}(X - e_i))) + V_{t+1}(X)$$

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$$\gamma_{i,t+1}$$

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$$\hat{V}_t(X) =$$

The Roll-Out Policy

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$$\hat{V}_t(X) = \sum_{i \in \hat{A}_t} \pi(i, \hat{A}_t) \cdot \Big(r_i - (\hat{V}_{t+1}(X) - \hat{V}_{t+1}(X - e_i))\Big) + \hat{V}_{t+1}(X)$$

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$$\hat{A}_t = \operatorname*{argmax}_{S \subseteq N(X)} \sum_{i \in S} \pi(i,S) \cdot (r_i - \gamma_{i,t+1})$$

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Claim: $\hat{V}_t(X) \geq H_t(X) = \sum_{i \in [n]} \gamma_{it} x_i$

Outline

Step 2: Give policy that earns $H_1(C)$

Step 3: Show that $2H_1(C) \geq Z^*$

The Choice-Based Deterministic LP

decision variables: $h_t(S)$ - fraction of time S if offered in period t

$$Z^* = \max \sum_{t\in[T]} \sum_{S\subseteq[n]} h_t(S) \cdot \sum_{i\in S} r_i \pi(i,S)$$

α_i (1) $\sum_{t\in[T]} \sum_{S\subseteq[n]} h_t(S) \cdot \pi(i,S) \leq C_i \quad \forall\, i \in [n]$

β_t (2) $\sum_{t\in[T]} h_t(S) \leq 1 \quad \forall\, t \in [T]$

Dual:

$$\min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S) \cdot (r_i - \alpha_i) + \sum_{i\in[n]} C_i \alpha_i \right\}$$

Theorem: A classical result is that $Z^* \geq \mathrm{OPT}$

(Later we will show that $2\mathrm{OPT} \geq Z^*$)

Liu Q, Van Ryzin G (2008) *On the choice-based linear programming model for network revenue management.* Manufacturing and Service Operations Management

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t \in [T]} \max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot (r_i - \alpha_i) + \sum_{i \in [n]} C_i \alpha_i \right\}$$

Claim: $2H_1(C) \geq Z^*$

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Claim: $2H_1(C) \geq Z^*$

Proof:

$$Z^* \leq \sum_{t \in [T]} \max_{S \subseteq [n]} \sum_{i \in S} \pi(i, S) \cdot (r_i - \gamma_{i1}) + \sum_{i \in [n]} C_i \gamma_{i1}$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

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$$= H_1(C)$$

The Choice-Based Deterministic LP

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Two key properties:

(i) $\gamma_{it} \geq \gamma_{i,t+1}$

(ii) $c_i\cdot(\gamma_{it}-\gamma_{i,t+1}) = \pi(i,A_t)\cdot(r_i-\gamma_{i,t+1})$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

$$\boxed{\textbf{Claim: } 2H_1(C) \geq Z^*}$$

Proof:

$$\begin{aligned} Z^* &\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \gamma_{i1}) + \sum_{i\in[n]} C_i\gamma_{i1} \\ &\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \gamma_{i,t+1}) + H_1(C) \end{aligned}$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t \in [T]} \max_{S \subseteq [n]} \sum_{i \in S} \pi(i,S) \cdot (r_i - \alpha_i) + \sum_{i \in [n]} C_i \alpha_i \right\}$$

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$$\leq \sum_{t \in [T]} \max_{S \subseteq [n]} \sum_{i \in S} \pi(i,S) \cdot (r_i - \gamma_{i,t+1}) + H_1(C)$$

$$A_t = \operatorname*{argmax}_{A \subseteq [n]} \sum_{i \in A} \pi(i,A) \cdot (r_i - \gamma_{i,t+1})$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

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$$= \sum_{t\in[T]} \sum_{i\in A_t} \pi(i,A_t)\cdot(r_i - \gamma_{i,t+1}) + H_1(C)$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

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Z^* &\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\gamma_{i1}) + \sum_{i\in[n]} C_i\gamma_{i1} \\
&\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\gamma_{i,t+1}) + H_1(C) \\
&= \sum_{t\in[T]} \sum_{i\in A_t} \pi(i,A_t)\cdot(r_i-\gamma_{i,t+1}) + H_1(C) \\
&= \sum_{t\in[T]} \sum_{i\in A_t} c_i\cdot(\gamma_{it}-\gamma_{i,t+1}) + H_1(C)
\end{aligned}$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

Claim: $2H_1(C) \geq Z^*$

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$$\begin{aligned}
Z^* &\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\gamma_{i1}) + \sum_{i\in[n]} C_i\gamma_{i1} \\
&\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i-\gamma_{i,t+1}) + H_1(C) \\
&= \sum_{t\in[T]} \sum_{i\in A_t} \pi(i,A_t)\cdot(r_i-\gamma_{i,t+1}) + H_1(C) \\
&= \sum_{t\in[T]} \sum_{i\in A_t} c_i\cdot(\gamma_{it}-\gamma_{i,t+1}) + H_1(C) \\
&\leq \sum_{t\in[T]} \sum_{i\in[n]} c_i\cdot(\gamma_{it}-\gamma_{i,t+1}) + H_1(C)
\end{aligned}$$

The Choice-Based Deterministic LP

$$Z^* = \min_{\alpha_i} \left\{ \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \alpha_i) + \sum_{i\in[n]} C_i\alpha_i \right\}$$

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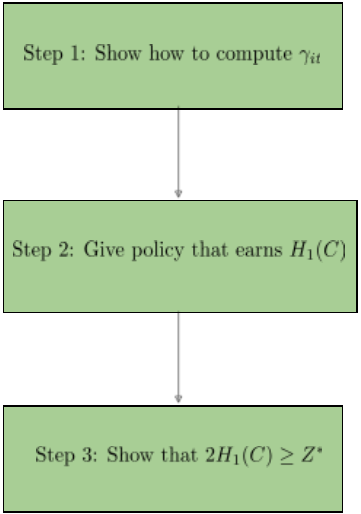
$$\leq \sum_{t\in[T]} \max_{S\subseteq[n]} \sum_{i\in S} \pi(i,S)\cdot(r_i - \gamma_{i,t+1}) + H_1(C)$$

$$= \sum_{t\in[T]} \sum_{i\in A_t} \pi(i,A_t)\cdot(r_i - \gamma_{i,t+1}) + H_1(C)$$

$$= \sum_{t\in[T]} \sum_{i\in A_t} c_i\cdot(\gamma_{it} - \gamma_{i,t+1}) + H_1(C)$$

$$\leq \sum_{t\in[T]} \sum_{i\in[n]} c_i\cdot(\gamma_{it} - \gamma_{i,t+1}) + H_1(C) \quad = \sum_{i\in[n]} c_i\gamma_{i1} + H_1(C)$$

Outline


Step 1: Show how to compute γ_{it}
Step 2: Give policy that earns $H_1(C)$
Step 3: Show that $2H_1(C) \geq Z^*$

Related Literature

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